

Sparse Interpretability of Audio Token Representations in Spoken Language Models

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Abstract

Spoken language models (SLMs) map speech and text into a shared representation space for unified sequence modeling, but the extent to which projected audio embeddings encode semantic versus acoustic information remains unclear. This study analyzes the internal structure of Qwen2-Audio’s audio representations using sparse autoencoders (SAEs) trained on frame-level audio tokens. The resulting features exhibit strong alignment with phonemes or phonetic categories, with some encoding context-dependent patterns such as word position. Apparent word-level meaning is explained by shared acoustic properties, not lexical semantics. Sparse features remain robust across changes in dimensionality and initialization, and exhibit consistent sparsity.

Index Terms: spoken language models, speech representation learning, sparse auto encoder

1. Introduction

Spoken language models (SLMs) have recently emerged as a promising framework for unified speech and text processing and achieve strong performance across a range of tasks, including automatic speech recognition, speech translation, and instruction following. Although its performance on speech-driven tasks is impressive, the inner workings of the representations it learns, particularly those that bridge acoustic and linguistic modalities, are still not well understood.

A key characteristic of these models is their ability to map both speech and text to a shared representation space for sequence modeling. Typically, a speech encoder transforms the input waveform into frame-level acoustic features and projected into token-level embeddings compatible with text. These embeddings are subsequently processed by a unified sequence model, often a decoder-only language model, which treats both modalities within the same generative architecture.

This design implicitly assumes that speech representations can be mapped into a semantic space similar to that of text, enabling the model to interpret speech and text within a shared framework. However, it remains unclear whether audio and text representations exhibit a similar internal structure or whether they fundamentally differ in the way they encode and organize information. A key open question is to what extent these projected audio representations encode semantic content rather than low-level acoustic features.

To address this question, we performed a systematic analysis of the internal structure of audio representations in Qwen2-Audio[1], a state-of-the-art speech-aware LLM. We focus on the pre-language model stage: specifically, the output of the model’s audio encoder.

To probe the structure and interpretability of these audio embeddings, we train sparse autoencoders (SAEs) as Figure 1 to

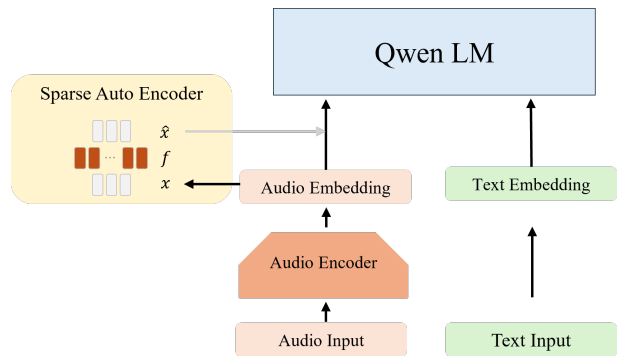


Figure 1: *Analysis Framework.*

learn overcomplete, sparse representations that expose latent dimensions of variation in the data. Prior work has demonstrated that SAEs can extract mono-semantic features in language models, where individual features consistently encode discrete semantic attributes such as emotions, entities, or syntactic roles. We apply this approach to Qwen2-Audio to examine whether similarly interpretable units emerge from audio token representations.

Our primary objective is to evaluate whether the model’s audio embeddings encode linguistic semantics rather than purely low-level acoustic properties. Gaining insight into these representations is essential for improving interpretability and guiding the design of more effective spoken language models.

2. Related Work

2.1. Representation Analysis in Speech Models

Understanding the information encoded in the internal representations of speech models has been a long-standing research focus. A common approach is *probing*, which involves training lightweight classifiers to decode attributes such as phonemes, speaker identity, emotion, or prosody from hidden states of models like HuBERT and Wav2Vec 2.0. Other studies employ metrics such as mutual information (MI) and canonical correlation analysis (CCA) [2] to examine how representations evolve across layers. These analyses have revealed that different types of information (e.g., acoustic, phonetic, semantic) emerge at different stages within the model. However, these methods offer only limited insight into the internal organization of the representation space and do not yield a structured or interpretable view of how such information is encoded.

2.2. Dictionary Learning and Sparse Autoencoders

Interpretable representation learning has gained increasing attention as deep models grow in complexity. Sparse autoencoders (SAEs), which impose sparsity constraints on their latent representations, have emerged as a powerful tool for uncovering structured and interpretable features. Notably, recent work by Anthropic [3] demonstrated that applying sparse dictionary learning to residual stream activations in transformer models can yield mono-semantic features—that is, individual features that consistently activate for a single, human-interpretable concept such as “Latin” or “DNA sequence.” They further explored the superposition problem, in which distributed representations entangle multiple concepts, and argued that sparse methods can help disentangle them.

In contrast to prior approaches that rely on supervised probing and focus primarily on textual transformers, we apply unsupervised sparse methods to directly characterize and compare the internal structure of audio and text token representations. We investigate whether similar monosemantic features arise in both modalities, and how their abstraction level, alignment, and compositional properties differ. To our knowledge, this work presents the first application of sparse interpretability techniques to a large-scale multimodal speech-language model[4].

3. Methods

This section outlines the methodology used to analyze the internal audio and text representations of Qwen2-Audio.

3.1. Feature Extraction

Frame-level audio token embeddings are extracted from Qwen2-Audio’s multi-modal projector. The audio signal is first processed by the model’s speech encoder to obtain frame-level acoustic features, which are subsequently projected to a token-level embedding space through a modality-specific adapter. To enable semantic evaluation of the learned features, phoneme and word-level alignments are obtained using the Montreal Forced Aligner (MFA), producing time-aligned labels for each audio frame. These labels are associated with the extracted audio embeddings for further analysis.

3.2. Sparse Autoencoder Architecture

To uncover latent structure in the audio token representations, Sparse autoencoders (SAEs) are trained on the audio token embeddings. SAE is implemented as an overcomplete, single-layer multilayer perceptron (MLP) with ReLU non-linearity. The encoder and decoder weights are untied, following previous findings that this improves model capacity and feature diversity.

Let the input embedding dimension be n , and the hidden (bottleneck) dimension be m , where $m \geq n$. Denote the encoder weights as $W_e \in \mathbb{R}^{m \times n}$, decoder weights as $W_d \in \mathbb{R}^{n \times m}$ (with decoder columns normalized to unit norm), and biases $b_e \in \mathbb{R}^m$, $b_d \in \mathbb{R}^n$. The forward computation proceeds as:

$$\begin{aligned} \bar{x} &= x - b_d \quad (\text{pre-encoder bias}) & (1) \\ f &= \text{ReLU}(W_e \bar{x} + b_e) \quad (\text{sparse hidden activation}) & (2) \\ \hat{x} &= W_d f + b_d \quad (\text{reconstruction}) & (3) \end{aligned}$$

Given a dataset $X = \{x^{(1)}, x^{(2)}, \dots, x^{(N)}\}$, the training ob-

jective is:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|x^{(i)} - \hat{x}^{(i)}\|_2^2 + \lambda \|f^{(i)}\|_1 \quad (4)$$

where the first term denotes the mean squared reconstruction loss and the second term imposes an ℓ_1 sparsity penalty on the hidden activations $f^{(i)}$. The hyperparameter λ controls the strength of sparsity regularization. Top- k sparsification may optionally be applied to enforce a fixed number of nonzero activations per sample.

Training is conducted using the Adam optimizer, with Kaiming Uniform initialization applied to both encoder and decoder weights. To encourage numerical stability and interpretability, decoder weights are re-normalized to have unit column norms after each update. Early stopping based on validation loss is employed to prevent overfitting. To address the issue of features that remain inactive due to enforced sparsity, an optional resampling mechanism on is introduced. The parameter of these “dead” neurons with persistently low activation frequencies are re-initialized, thereby improving overall dictionary coverage and promoting more effective utilization of the latent space.

4. Experiments

4.1. Setup

This part explains our details of experiment settings.

Data. Experiments are conducted on the LibriSpeech train-clean-100 subset, which provides English read speech along with time-aligned transcriptions. To obtain frame-level alignment between audio and text, forced alignment is performed using Montreal Forced Aligner (MFA), resulting in phoneme and word-level labels for each token in TextGrid format.

Audio Tokens. The analysis focuses on the multi-modal projector outputs of Qwen2-Audio-7B. Each input utterance is first converted into a 128-channel log mel-spectrogram and passed through a convolutional and transformer-based audio encoder. Temporal resolution is reduced by a factor of 4, resulting in one frame every 40 ms. The encoder outputs frame-level vectors of shape $T \times 1280$, which are then linearly projected into a 4096-dimensional token space by the multi-modal projector, yielding a sequence of audio tokens of shape $T \times 4096$ as input to the Sparse Autoencoder.

4.2. Feature Interpretability Analysis

4.2.1. Phonetic Level Meaning

Our analysis reveals a consistent and interpretable alignment between sparse features and phoneme identities.

Vowel-Feature Activation Patterns. To quantify the phoneme-level selectivity of sparse units, Figure 2 presents a heatmap showing the mean activation of each sparse feature (column) across frames aligned with individual vowel phonemes (row). Each cell represents the average activation of a given feature on a particular vowel, aggregated across all its occurrences in the dataset.

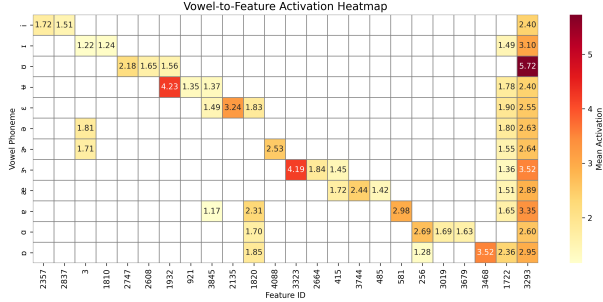


Figure 2: Vow to feature activation heatmap.

The diagonal pattern observed in the heatmap suggests that many sparse features are highly selective to specific vowels. For instance, Feature 2608 shows the strongest activation on /æ/. Feature 1820 is activated by /ε/ and /a/, which has a similar spoken.

These observations provide additional evidence that sparse auto encoding induces interpretable and phonologically grounded representations. It also indicates that the learned sparse features are highly mono-semantic and interpretable and serve as a meaningful base for the token space.

However, several features such as 3293 and 1722 exhibit high mean activations across multiple vowels, suggesting that they may not encode individual phonemes but instead capture broader articulatory or prosodic patterns. This indicates that while many features are phonetically selective, some encode more abstract or distributed acoustic properties.

Coverage and Distribution. To assess whether sparse codes encode phoneme class-level distinctions, a UMAP projection is applied to the frame-level activation vectors from the sparse autoencoder. In Figure 3, each point represents a single frame and is colored according to the aligned phoneme category, including front, central, and back vowels, as well as voiced and voiceless consonants.

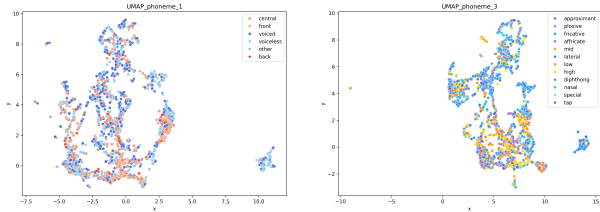


Figure 3: UMAP projection colored by phoneme categories.

The projection reveals structured organization of phoneme categories. Vowels, particularly those labeled front, central, and back, tend to form localized regions along the boundary of the embedding space. Voiced and voiceless consonants are interspersed in denser clusters toward the interior and right-hand region, with a distinct subregion (rightmost cluster) composed predominantly of voiceless consonants, suggesting high separability of certain articulatory classes.

Although some category boundaries appear fuzzy—particularly between central vowels and voiced consonants—the overall clustering pattern indicates that the sparse autoencoder captures phonetic structure in a geometrically meaningful way.

Context Sensitivity. Same phoneme activates different features depending on its position within the word. As illustrated

in Table 1, the same phoneme can activate different features depending on its location within a word. For instance, Feature 2570 is consistently activated by the stop consonant /t/ when it appears at the word-final position, such as in “but” and “at.” In contrast, Feature 2990 responds to the voiceless fricative /θ/ primarily in word-initial contexts. These observations suggest that the learned features not only encode phonetic identity but also capture fine-grained positional information, indicating a degree of contextual abstraction within the representation space.

Table 1: Examples of features with phonetic-level meanings.

Feature ID	Phoneme	Category	Example Word(s)
2570	/t/ (word-final)	Stop	but, at
2990	/θ/ (word-initial)	Voiceless Stop	these
3293	/I/ (word-middle)	Short front vowel	his, him
3055	/i:/ (word-final)	Long vowel	he, she

4.2.2. Word Level Meaning

Although certain sparse features seem to correspond to specific word categories such as pronouns, modal verbs, or prepositions, further analysis in Table 2 shows that these patterns are primarily driven by underlying acoustic similarities rather than true lexical semantics. For instance, pronouns like his, him, and her typically contain short central vowels such as /I/ or /ə/. Modal verbs like would and could often include stop consonants such as /d/ or /t/. Prepositions such as to, for, and of consistently involve reduced vowels and soft consonants. These findings indicate that the sparse autoencoder captures low-level phonetic regularities that are correlated with surface-level word categories, rather than learning abstract semantic representations.

Table 2: Examples of features with word-level meanings.

Feature	Meaning	Examples	Acoustic Pattern
3293	Pronouns	his, him, her	Short Vowel (/I/, /ə/)
2621	Modal Verbs	would, could	Stop consonant(/d/, /t/)
3673	Prepositions	to, for, of, at	neural vows + soft consonant

4.3. Feature Robustness Analysis

To quantify the consistency of learned features across different training conditions, we use Activation Similarity, a metric that compares the alignment of sparse units between two models. Specifically, given two trained autoencoders A and B , we compute the Pearson correlation between each feature vector f_i^A from model A and all feature vectors f_j^B from model B . For each i , we take the maximum correlation over j , and report the median across all $i \in \{1, \dots, d\}$:

$$\text{Activation Similarity} = \text{Median}_{i=1}^d \left(\max_{j=1}^d \text{corr}(f_i^A, f_j^B) \right)$$

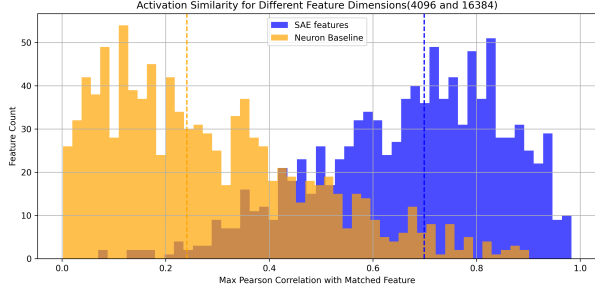


Figure 4: *Activation similarity between different 4096 and 16384 Feature Dimensions.*

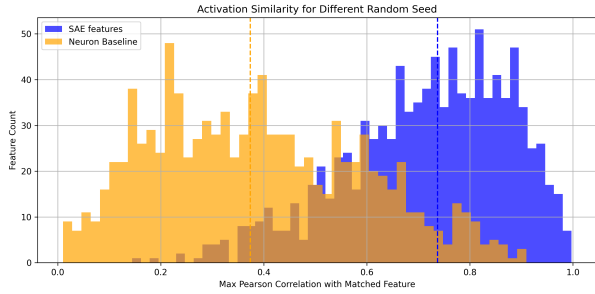


Figure 5: *Activation similarity between different random seeds.*

Using this metric, we compare activation matrices constructed from sparse features trained with a Sparse Autoencoder to those derived from dense representations (raw neuron activations without sparsity constraints). As shown in Figure 4 and Figure 5, sparse features exhibit significantly higher activation similarity across different model initializations (random seeds) and across varying feature dimensionalities (e.g., 4096 vs. 16384), whereas dense neurons are far more variable under these changes. This indicates that sparse features are more robust and semantically consistent, suggesting that the SAE extracts more stable and interpretable structure from the same input space.

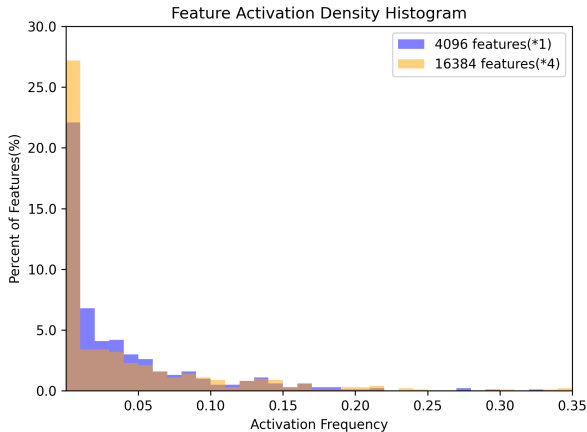


Figure 6: *Feature sparsity*

In addition to robustness, we observe from Figure 6 that

the learned SAE features consistently maintain sparse activation patterns across different model capacities. Regardless of whether the feature dimensionality is set to 4096 or 16384, the majority of features are activated by only a small fraction of inputs—typically less than 5%. Only a small subset of features exhibit broader activation, firing on more than 20% of inputs. This distribution indicates that the SAE naturally induces a high degree of sparsity, enhances interpretability and minimizes feature overlap, aligning with the goal of discovering monosemantic representations.

5. Conclusions

Our analysis shows that the audio encoder in Qwen2-Audio captures a well-organized hierarchy of phonetic information. Many sparse features learned by the Sparse Autoencoder align closely with specific phonemes or phonetic categories, and some are sensitive to contextual factors such as position within a word or surrounding phonetic environment. This suggests that the encoder is able to represent fine-grained acoustic details in a structured way.

Some features also show consistent activation on common function words. However, further analysis indicates that this is not due to word meaning, but rather to shared acoustic properties such as short duration, pitch pattern, or vowel quality. These patterns imply that the apparent word-level selectivity is likely a result of surface-level acoustic regularities, rather than deeper lexical or semantic encoding.

Finally, we find that the learned sparse features are robust. They remain stable across different settings, including changes in feature dimension and random seed. Sparsity is also consistently maintained, with most features activating on only a small fraction of inputs. This supports the idea that the SAE captures meaningful and reusable structure in the audio representations.

6. Future Work

Beyond the current analysis, several promising directions remain open for future exploration.

First, future work could investigate how sparse representations compare with other interpretability methods, such as layerwise analyses based on canonical correlation analysis (CCA) or mutual information (MI), as well as techniques like probing and causal interventions.

Another promising direction is to extend sparse analysis to the text inputs and therefore the language model part after the encoder, particularly the attention heads. This may provide deeper insight into how text and audio information are fused and transformed across layers, thereby enhancing our understanding of joint speech-text modeling in large-scale architectures.

In addition, scaling up the dataset in both size and diversity would allow for the emergence and identification of a broader range of high-level acoustic features, such as emotion, intonation, prosody, and background sound. These attributes are typically absent in pure text inputs, raising the question of how large language models internalize and make use of such rich paralinguistic information when exposed to audio input.

Finally, sparse features could be further explored for their utility in downstream tasks. This includes using sparse codes as interpretable and compact inputs for tasks such as emotion recognition, keyword spotting, or speaker verification. Moreover, by intervening on specific sparse units, it may be possible to steer model behavior in controlled ways—for instance, altering the emotional tone or formality of the generated

7. References

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